6.5 Differentiation from Axioms

WORKSHEET

The question

STEP II 1992, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s2-q4

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself Rule (i) is the product rule. How could the product rule tell you anything about a constant? (Think of a constant as a product.)

STAGE 1 Constants

Covers: constants

Ask yourself Rule (i) is the product rule. How could it tell you anything about a constant? Think of 1 as 1×1.

Pause and try Apply rule (i) with $f = g = 1$. Then use rule (iv) for any other constant.

STAGE 2 Powers

Covers: the powers x^2 and x^3

Ask yourself Write $x^2 = x \cdot x$ and $x^3 = x^2 \cdot x$, and use rules (i) and (iii).

Pause and try Prove by induction that $\Diamond x^n = nx^{n-1}$.

STAGE 3 All polynomials

Covers: any polynomial

Ask yourself Write a general polynomial as $a_0 + a_1x + \cdots + a_mx^m$, and use rules (ii) and (iv).

What's really going on

Derivations

An operation D on an algebraic system that is additive and satisfies the product rule $D(fg) = fD(g) + gD(f)$ is called a *derivation*. The question shows that on polynomials, a derivation that kills constants is completely determined by $D(x)$: in fact $D(h) = h'(x)\,D(x)$, so the only freedom is the choice of the polynomial $D(x)$. Rule (iii) fixes $D(x) = 1$. Derivations appear across mathematics: in geometry, a *vector field* on a surface can be defined as a derivation on the functions on that surface (it differentiates functions in the direction of the field); in algebra, derivations of a structure measure its "infinitesimal symmetries"; in physics, the commutator $[A, B] = AB - BA$ of quantum mechanics is a derivation in each argument.

Calculus without limits

The algebraic derivative works in settings where limits make no sense at all. Over a finite field, such as arithmetic modulo a prime p, there are no limits, but the formal derivative of a polynomial is defined by the same rules and still detects repeated roots: a polynomial f has a repeated root exactly when f and f' have a common factor. That is how one shows, for example, that $x^n - 1$ has no repeated roots when $p \nmid n$, a fact needed to define the cyclotomic polynomials of Section 6.1 over finite fields. The same idea is used in coding theory and cryptography.

Automatic differentiation

Here is a modern application. Invent a number ε with $\varepsilon^2 = 0$ but $\varepsilon \neq 0$, and do arithmetic with *dual numbers* $a + b\varepsilon$. Then $(a + b\varepsilon)(c + d\varepsilon) = ac + (ad + bc)\varepsilon$: the ε-part

follows exactly the product rule. So for any polynomial,

$$f(x+\varepsilon) = f(x) + f'(x)\,\varepsilon,$$

and evaluating f at the dual number $x+\varepsilon$ computes the value and the derivative at the same time, exactly, with no limits and no rounding error from small differences (Figure 6.5.1). This is called *forward-mode automatic differentiation*. Its cousin, reverse mode, is the algorithm (*backpropagation*) used to train neural networks with billions of parameters: every time such a system learns, it is applying the product and chain rules, as derivations, step by step.

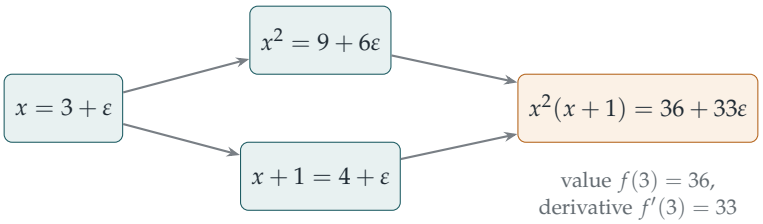


Figure 6.5.1 Automatic differentiation with dual numbers ($\varepsilon^2 = 0$) for $f(x) = x^2(x+1)$ at $x = 3$. Each step follows the rules of the question: $(3+\varepsilon)^2 = 9 + 6\varepsilon$, and $(9+6\varepsilon)(4+\varepsilon) = 36 + (9+24)\varepsilon$. The coefficient of ε is $f'(3) = 3 \cdot 9 + 2 \cdot 3 = 33$, computed exactly.

Try this next

1. **A different derivation.** Suppose $\Diamond$ satisfies rules (i), (ii) and (iv), but $\Diamond x = x$ instead of 1. Show that $\Diamond \mathrm{h}(x) = x\,\mathrm{h}'(x)$. What does $\Diamond$ do to x^n?

2. **The quotient rule.** Extend $\Diamond$ to rational functions f/g, assuming rule (i) still holds. By applying rule (i) to $\mathrm{g} \cdot \frac{\mathrm{f}}{\mathrm{g}} = \mathrm{f}$, derive the quotient rule.

3. **Repeated roots.** Show that if $(x-a)^2$ divides a polynomial f, then $x-a$ divides f'. Use this to show that $x^n - 1$ has no repeated roots.

4. **Dual numbers.** Use dual numbers to compute $f(2)$ and $f'(2)$ for $f(x) = (x^2+1)^3$, and check with the chain rule.

Further reading

- **Peter Cameron, *Introduction to Algebra*** (Oxford University Press). Rings, polynomials and fields at first-year level, with the formal derivative.
- **Andreas Griewank & Andrea Walther, *Evaluating Derivatives*** (SIAM). The standard reference on automatic differentiation. Advanced.
- **Michael Nielsen, *Neural Networks and Deep Learning*** (free online). Chapter 2 explains backpropagation clearly.

The full guided solution is in Section 6.5:
https://beyondstep.keithmcnulty.org/ch06/derivations.html